

# VAYUSETU: Red-Team Security & Robustness Audit

Failure Mode Analysis, Fail-Safe Fallbacks, & Security Compliance

## 1. Audit Purpose & Scope

This document outlines the security, resilience, and robustness audit performed on VAYUSETU to identify failure vectors, data anomalies, and boundary violations, ensuring the system can support critical early warnings for government agencies without crashing during disasters.

## 2. Failure Scenarios and Mitigations

- **Scenario 1: Ground Station / Satellite API Failure**

*Risk:* Raw INSAT or IMD servers go offline.

*Mitigation:* The backend automatically falls back to climatological monthly means for the district, maintaining dashboard updates while logging an infrastructure warning in the telemetry stream.

- **Scenario 2: Data Freshness Stale**

*Risk:* Live ingestion pipeline stalls.

*Mitigation:* The freshness monitor checks timestamps. If a feed is older than 24 hours, its status shifts from LIVE to STALE, triggering visual warnings on the UI status bar.

- **Scenario 3: Model Drift or Anomaly**

*Risk:* Extreme weather causes out-of-distribution inputs, leading to bad AI predictions.

*Mitigation:* The drift detector flags when incoming distributions differ significantly from training data, advising a manual model retraining click and displaying wider uncertainty bounds (lower confidence).

- **Scenario 4: System Offline / Zero Connectivity**

*Risk:* Operator loses network access in the field.

*Mitigation:* The dashboard integrates a 'Demo Mode' switch, loading cached spatial-temporal grids and mock-running equations locally in the browser/sqlite client.

## 3. Technical Robustness Checklists

1. Checked sanitization of district search parameters to prevent SQL injections.
2. Validated bounds of what-if sliders to prevent out-of-range numerical overflows in the Saint-Venant solver.
3. Configured Leaflet bounds to avoid infinite map repetition and horizontal scrolling glitches.